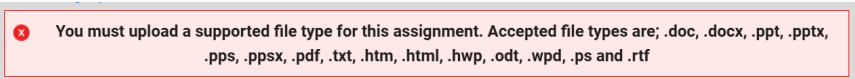




FIT1047 Introduction to computer systems, networks and security – S2 2025

Assignment 1 – Numbers, Encodings and Boolean Logic

Purpose	In this assignment, you will demonstrate your knowledge of number systems, character encodings and that you can construct and simplify Boolean formulas and circuits. The assignment relates to Unit Learning Outcomes 1 and 2.
Your task	Part 1: Submit your reflections (Weeks 1 - 3). See details below. Part 2: Complete the individual tasks as detailed in the instructions below. You need to submit a document that details your workings, as well as Logisim files for the resulting circuits. Part 3: Complete the in-class test. In-person attendance required. <u>You must come to your allocated Applied session (for Australian cohort) / Workshop session (for Malaysian cohort) in week 4 in-person.</u> See Moodle and announcements for more details.
Value	15% of your total marks for the unit. See below for how marks and grades are determined.
Word Limit	No overall word limit (see instructions for word limits of individual tasks)
Due Date	Part 1 and 2: 11:55 PM Friday 15 August 2025 Part 3: During your allocated applied session (Australian cohort) / workshop session (Malaysian cohort) in Week 4 (18-22 August 2025)
Submission	• Via Moodle Assignment Submission. • Turnitin will be used for similarity checking of all submissions. Please ignore the following error message from Turnitin for zip files or other non-document files:  • DRAFT upload confirmation email from Turnitin is not a submission. You must click the submit button to accept terms and conditions in Moodle. DRAFT submissions will not be assessed. • Do not use GDrive/OneDrive/etc. shared links for submission. Please export the document(s) as a PDF and then upload it in Moodle for grading. • Once the submission is confirmed, any requests to revert it back to DRAFT will not be accepted. Also, any incorrect, corrupted, empty or wrong file type submission will not be assessed. Please check carefully before confirming your submission.

	• This is an individual assignment (group work is not permitted). • Handwritten work, including digital work created by touchscreen/digitizer technologies, will not be assessed. • Logisim circuits will be assessed using version 3.9.0 (link is in Moodle). • In this assessment, you must not use generative artificial intelligence (AI) to generate any materials or content in relation to the assessment task.
Assessment Criteria	Marks are awarded for the <i>correctness</i> of the calculations, the <i>explanations</i> of how the tasks were solved, and the documentation of <i>test cases</i> where required. The marking rubric in Moodle shows an individual marks breakdown. If no or insufficient reflections are submitted (Part 1), the overall marks for this assignment are capped at 30 (i.e., 50% of the overall mark).
Late Penalties	• 5% deduction per calendar day or part thereof for up to one week • Submissions more than 7 calendar days after the due date will receive a mark of zero (0) and no assessment feedback will be provided.
Support Resources	See Moodle Assessment page
Feedback	Feedback will be provided on student work via: • general cohort performance • specific student feedback ten working days post submission

INSTRUCTIONS

This assignment has three parts. Make sure you read the instructions carefully.

For **Part 1**, collect your **reflections** for Weeks 1 - 3 from each week's Ed Lesson and create a single PDF document. You can simply copy/paste your reflections, but please add headings for each week. Please refer to the [template](#) as a reference of the structure of your reflections. While it is optional to follow the template, you are required to have your reflections *specific* to the topics covered in the corresponding weeks. The word count of reflection must be more than 100 words (excluding the headings provided in the template) in each week. **Submit your PDF through the Moodle Assignment 1 Part 1 activity.**

For **Part 2**, you need to submit two files through the Moodle **Assignment 1 Part 2** activity:

- 1) A **PDF** document with the answers to the questions and your workings.
- 2) A separate **.zip archive with the Logisim files** (do NOT save as PDF file for your Logisim files) along with your **individualised assignment specification**.

Part 3 is an **in-class test** during your allocated Applied session (for Australian cohort) / Workshop session (for Malaysian cohort) in Week 4. Instructions will be available separately.

How are marks and grades determined?

Part 2 and Part 3 are worth 30 marks each. The overall mark is the sum of the two individual marks. The assignment is worth 15% of the unit's marks.

If no meaningful or insufficient reflections are submitted for Part 1, the overall mark will be:

- 30 if the sum of Part 2 and Part 3 is greater than or equal to 30;
- the sum of Part 2 and Part 3, otherwise.

Detailed Instructions for Part 2: Boolean Algebra (30 marks total)

Follow the link on Moodle to access your **personalised truth table** for this task.

Important: Your truth table is different from the one other students are working on. Only download the file while you are correctly logged into Moodle with your own student account. Submitting work based on an incorrect truth table will be investigated as a potential breach of academic integrity.

The truth table you download describes a Boolean function with four input values x_1, x_2, x_3, x_4 and two output values z_1, z_2 .

The main result of this part will be a logical circuit correctly implementing this Boolean function in the Logisim simulator.

Each task below needs to be documented and explained.

Task 2.1: Boolean Algebra Expressions (10 marks)

Write the Boolean function as Boolean algebra terms using the Sum of Product (SOP) form. First, think about how to deal with the two outputs. Then, describe each single row, in the truth table, in terms of Boolean algebra. Finally, combine the terms for single rows into larger terms.

Briefly explain these steps for your particular truth table (e.g., explain for one particular row how you come up with the Boolean terms for that row, and then explain how you combine all rows). This explanation should be no more than a few sentences.

Also, write the Boolean function as Boolean algebra terms using the Product of Sum (POS) form with a brief explanation of steps (e.g. how you combine all rows). Note that details of POS are not covered in lectures/workshops/applied classes and you are required to find out how it works and come up with the Boolean function.

Notation: Use the following symbols and notation for writing Boolean algebra expressions. Variables are upper-case (e.g., X_1, Z_2). Boolean AND is written without a symbol, e.g., X_1X_2 . Boolean OR is written with the + symbol, e.g. $X_1 + X_2$. Negation is written using an overline, e.g., $\overline{X_1}$. **Important:** when writing terms like NOT X_1 AND NOT X_2 , there must be a clear gap in the overlines, e.g., $\overline{X_1} \overline{X_2}$. Tip: **you can use the equation function in Word or Google Docs to create overlines**.

Task 2.2: Logical circuit in Logisim (10 marks)

Model the resulting Boolean terms (using SOP form) from Task 2.1 in a single Logisim circuit, using only the basic gates AND, OR, NOT. You can use gates with more than two inputs. **Check the [template on the last page](#) for how to structure your circuit.**

Briefly explain your construction (as for Task 2.1, a short explanation is enough).

Test your circuit using values from the truth table and document at least 3 test cases. You can take screenshots of your Logisim window to document the tests. For each test case, state clearly the expected *input* and *output* values, in terms of a table or labels on the screenshot.

Task 2.3: Optimised circuit (10 marks)

The goal of this task is to find a **minimal** circuit using only AND, OR, and NOT gates. Based on the truth table and Boolean algebra terms of SOP from Task 2.1, optimise the function **using Karnaugh maps**.

You will need to create two Karnaugh maps, one for each output. Your documentation must show

- 1) the maps,
- 2) the groups found in the maps
- 3) the reduced Boolean functions derived from the maps and how the maps relate to terms in the optimised Boolean functions.

Then, **use Logisim to create a minimal circuit**, using only AND, OR, and NOT gates.

Test your circuit using values from the truth table and document at least 3 test cases. You can take screenshots of your Logisim window to document the tests. For each test case, state clearly the expected *input* and *output* values, in terms of a table or labels on the screenshot.

Submission Checklist

Altogether, you should submit the following files:

- **Part 1**
 - **1 PDF file for your reflections (Weeks 1 - 3)**
- **Part 2**
 - **1 PDF file - your answers to the questions**
 - **1 ZIP file containing the following files:**
 - **2 Logisim files**
 - **1 PDF file for your individual truth table**

Please check CAREFULLY before confirming your submissions.

Any requests to revert the submission back to DRAFT will not be accepted.

DRAFT submissions will not be assessed.

Also, any incorrect, corrupted, empty or wrong file type submission will not be assessed.

Logisim Template

Structure your circuits as shown in the Logisim diagram below (keep both outputs in the same circuit). This template is available on Moodle.

